Compute Dilution in Budgeted Online RGB-D Gaussian Splatting

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Abstract

Online RGB-D 3D Gaussian Splatting (3DGS) spends a fixed per-frame optimization budget while its map keeps growing. We study ***compute dilution****: as the map grows, established Gaussians receive a shrinking share of the updates. Interventions on two TUM scenes (150 frames) and two Replica scenes (99 and 119 frames), eight seeds each, and eleven further scenes (99 frames, three seeds) support this as a compute-allocation effect. Raising the optimization budget in proportion to map size, with the spawn allowance fixed, improves raw-error selection on all 15 scenes (median* $+2.70$ dB*, range* $+0.02$ *to* $+3.73$ dB*); with free map growth, PSNR rises monotonically with budget and tracks the fraction of mature Gaussians updated per frame (per-seed Spearman* $+1.00$ *in every scene). The usual remedy, throttling new Gaussians, is scene dependent: coverage throttling improves raw-error selection on the two main TUM scenes (*$+3.53$ *and* $+1.71$ dB*) but only on 5 of 15 scenes overall, and lowers it on Replica (*-0.68 *and* -0.87 dB *on the two scenes with full factorials). At equal measured time a spawn cap is the better use of compute on TUM and the worse on Replica, and with estimated poses the gains persist at the tracker's pose but nearly vanish when rendering from the true pose. On held-out frames the proportional-budget gains keep their sign but retain only* 47–76% *of their size. A throttled policy with temporal-influence selection reaches* 27.74 dB *on* `fr2_xyz`*,* 2.36 dB *above the strongest budgeted baseline and* 0.15 dB *below unconstrained optimization, with a* $3.2\times$ *smaller map, but it trails raw-error selection on 10 of 14 sequences, including all eight Replica scenes. The effect of coverage throttling comes almost entirely from its spawn cap, and a fixed spawn cap matches it on TUM; hard caps remove capacity on Replica, so map size is not isolated as the cause. On 500-frame TUM sequences the baseline degrades with horizon and the gain from the proportional budget grows (from* $+1.42$ *to* $+3.65$ *dB over the first 150 frames to* $+5.77$ *to* $+8.48$ *dB over frames 301–500). The* 15 ms *limit is modeled optimization time rather than end-to-end latency (about 10 FPS measured), and seeds measure repeatability on fixed trajectories, not generalization across scenes. Auxiliary analyses (counterfactual Gaussian utility, a learned utility model, contextual re-ranking, and a hole-rate policy router) are in the supplement.*

1. Introduction

3D Gaussian Splatting (3DGS) [6] underlies online RGB-D mapping systems such as SplaTAM [5], Point-SLAM [13], MonoGS [10], Photo-SLAM [4] and RTG-SLAM [12]. Deployed on robots and headsets, these systems have only milliseconds of optimization per streaming frame. At the same time, online pipelines add Gaussians in response to rendering error, so the map grows while the per-frame budget stays fixed; in our runs some maps exceed 100k primitives within 150 frames.

We hypothesize **compute dilution**: a growing map spreads a fixed optimization budget over more Gaussians and over newly spawned ones, so established (mature) Gaussians are updated less often and quality stagnates or degrades. We test this with interventions: a corrected-pose selector$\times$throttle factorial, optimization-budget controls (a fixed $\times 4$ budget, a budget proportional to map size, and a $\times 1/\times 2/\times 4$ dose-response) that hold the spawn allowance fixed, hard map caps, and per-frame logs of how the budget is allocated by Gaussian age, on two TUM and two Replica scenes, extended to 15 scenes for the proportional-budget and coverage contrasts. Raising the budget in proportion to map size improves PSNR on all 15 scenes (median $+2.70$ dB, range $+0.02$ to $+3.73$ dB), and with free growth the fraction of mature Gaussians updated per frame tracks quality in the dose-response. The standard remedy, throttling spawns, is not universal: it adds $+3.53$ and $+1.71$ dB on the two main TUM scenes but improves PSNR on only 5 of 15 scenes and lowers it on Replica. An ablation shows that the spawn cap, not the accompanying threshold change, carries the throttling effect, and that a fixed spawn cap matches coverage gating on TUM. Hard caps remove capacity on Replica, so we claim a compute-allocation mechanism under the tested protocol, not that map size is the unique cause.

Contributions. (1) A measurable formulation of compute dilution, with age-resolved allocation statistics (budget

share spent on young Gaussians, selection rate of mature Gaussians) logged per frame. (2) Interventional evidence on four scenes (two TUM, two Replica; eight seeds each): a 2×3 selector–throttle factorial, fixed and map-proportional budget controls, a budget dose-response, and map-cap controls; and a 15-scene analysis in which the proportional budget helps everywhere and coverage throttling in 5 scenes, with limits stated. (3) A simple coverage-throttled policy that reaches 27.74 dB on `fr2_xyz` within 0.15 dB of unconstrained optimization at a $3.2\times$ smaller map, together with a scene-by-scene account of where it fails (10 of 14 sequences, all Replica scenes), including a Replica factorial showing that throttling itself lowers PSNR there. (4) Auxiliary negative and descriptive results (counterfactual single-Gaussian utility, a learned utility model, contextual re-ranking, a hole-rate router), reported in the supplement with frozen artifacts.

2. Related Work

Online 3DGS SLAM and dense mapping. GS-SLAM [19] and SplaTAM [5] introduced differentiable 3D-Gaussian SLAM, with SplaTAM using silhouette-guided optimization; Gaussian-SLAM [21] adds a sub-map design and LoopSplat [23] adds loop closure by registering Gaussian submaps; Point-SLAM [13] uses dynamic neural point clouds; MonoGS [10] and Photo-SLAM [4] push toward real-time operation with loop closure; RTG-SLAM [12] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (reimplemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 5). **Scalable and efficient Gaussian mapping.** Recent systems address online map scalability through representation and update design. VTGaussian-SLAM [2] ties Gaussians to RGB-D views and avoids keeping every primitive learnable, targeting large scenes and memory use. SEGS-SLAM [18] improves structure and appearance consistency, while DyGS-SLAM [3] focuses on dynamic-scene localization and reconstruction. In parallel, SteepGS [17] derives compact density-control updates for 3DGS reconstruction. A recent online-mapping preprint studies geometry-aware initialization and densification under short optimization schedules [8]. These methods make clear that map representation, density control, and online optimization are coupled. Our study asks how growth rate and update policy should respond to coverage and observable scene statistics under a fixed per-frame budget and Gaussian representation. We do not claim density control or efficient Gaussian mapping as new in general. The per-scene SOTA numbers in Appendix Supp. Tab. S2 are protocol-mismatched context. **Density control for 3DGS.** Taming 3DGS [9] uses guided, constructive densification under resource limits; Mini-Splatting [1] targets a constrained Gaussian count; Pixel-GS [22] reweights the densification gradient by covered pixels; and 3DGS-MCMC [7] replaces heuristic densification and pruning with relocation. These methods address offline reconstruction. Our setting differs in that a per-frame optimization budget is shared between existing and newly spawned Gaussians. We do not benchmark against them, which is a limitation of the present comparison (Sec. 5). **Evaluation protocols in dense SLAM.** Published systems report full-sequence train-view rendering with global refinement at full resolution (Replica 33–42 dB for MonoGS/Point-SLAM/RTG-SLAM; TUM fr2_xyz 16–25 dB for MonoGS/SplaTAM/Photo-SLAM) plus full-trajectory ATE. Our protocol is deliberately narrower — budgeted online mapping on fixed horizons — so cross-paper numbers are contextual range, never superiority claims (Supp. Tab. S2). **Budgeted and utility-driven selection.** Error-only, error×influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers. Their ordering differs between the evaluated TUM and Replica scenes (Sec. 4). **Submodular maximization and knapsack allocation.** Budgeted set selection is related to monotone submodular maximization, for which approximation guarantees exist under explicit assumptions and appropriate algorithms [11, 16]. Those assumptions are not established for our measured reconstruction objective, and our fractional-knapsack ranker does not inherit those guarantees. The measured interactions vary by protocol, so we treat pointwise ranks as references and test map-growth control empirically.

3. Setting and Method

Budgeted mapping. Let $G_t = \{g_1, \ldots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t. Per frame, a scheduler chooses a subset $A_t \subseteq G_t$ to optimize,

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \tag{1}$$

where ΔQ is the reconstruction improvement, approximated at runtime by a selector score, and $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ is a decoupled cost model for K microsteps. The budget B_t is *modeled* optimization time. **Measuring dilution.** We log two age-resolved statistics per frame: the *young budget share*, the fraction of modeled update budget spent on Gaussians younger than ten frames, and the *mature selection rate*, the fraction of Gaussians at least ten frames old that are selected. Dilution predicts a falling mature selection rate as the map grows. **Selection.** Candidates are scored by

error × influence mass, optionally with temporal EMA filtering ($\beta = 0.90$), and packed by fractional-knapsack value density s_i/C_i under $B_t = 15\,\text{ms}$ with safety factor 1.10 (effective cap 13.6 ms). Each selected Gaussian receives $K = 5$ microsteps; the cost model is calibrated so modeled and wall-clock optimization time agree within measurement noise on the RTX 4050 backend. Exact Gaussian-to-pixel attribution costs 523.9 ms/frame, so we use a center-sampled projection, $\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4}\, e(\lfloor \mu_{2D,i} \rceil)$, that runs in 0.85 ms (614.5× faster) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. The scheduler totals 2.54 ms per frame. **Coverage throttling.** Coverage κ_t is the fraction of valid pixels with transmission < 0.5. The per-frame spawn cap ramps from 1.0 to 0.20 as κ_t goes from 0.80 to 0.90. A backlog limit exists but was inactive in the headline runs, because its count was only computed when warmup was enabled; the factorial evaluates it independently. Warmup pre-allocation is disabled and microsteps are fixed at $K = 5$.

4. Experiments

Protocol. TUM RGB-D [15] and Replica [14] sequences at 320 × 240, $B_t = 15\,\text{ms}$, seeds [42..49]; two-sided paired Wilcoxon tests, Holm correction and bootstrap CIs over per-seed trajectory summaries; production `gsplat` backend; ground-truth poses with tracking decoupled (a mapping benchmark, not SLAM ATE). With eight seeds the smallest attainable two-sided Wilcoxon p is 0.0078, and with fixed trajectories seeds measure repeatability only, so we report effect sizes with intervals and do not treat seed-level standardized effects as evidence of generalization. Sequences use the first 150 frames where available; the local Replica copies provide 99 evaluated frames for `office0` and 119 for `office1`. All results in Tabs. 1–4 come from one code revision (git tag `paper-rerun-1`) with the corrected evaluator: standard Gaussian-window SSIM and alpha-normalized rendered depth. Earlier bundles used a global-statistics SSIM and a depth evaluator that biased depth L1 toward zero in partially transparent pixels, so their SSIM and depth values are not comparable with these. Artifacts: `results/rerun_paper1/`. **Holdout protocol.** Because online mapping optimizes on the evaluated trajectory, train-view PSNR overstates generalization. We hold out every 8th frame render-only (never mapped, never selected) and report held-out effects in Tab. 8. This is a same-trajectory proxy, not independent novel-trajectory generalization.

Main result (Tab. 1; Supp. Figs. S4–S5). OURS reaches 27.74 ± 0.03 dB: +2.36 over the strongest budgeted baseline, +4.21 over the RTG-SLAM-policy re-implementation, and +8.89 over pass-through. These results evaluate the combined policy on this trajectory.

Selector × throttle factorial (Tab. 2). With selector fixed to raw error, coverage adds +3.53 [3.38, 3.70] dB on `fr2_xyz` and +1.71 [1.64, 1.80] dB on `fr1_xyz`. With temporal×influence fixed, its gains are +2.49 [2.40, 2.59] and +1.50 [1.48, 1.52] dB. Brackets are paired seed-bootstrap 95% CIs; all four coverage contrasts have two-sided paired Wilcoxon $p = .0078$. Temporal selection itself adds +0.09 dB with coverage on `fr2_xyz` and +0.54 dB on `fr1_xyz`; the interaction is negative (−1.04 and −0.22 dB), so selector and coverage effects are not additive. The independent backlog option adds +0.01/−0.02 dB on `fr2_xyz` and +0.08/+0.11 dB on `fr1_xyz` for error/temporal selection.

Dilution controls and direct allocation. Holding the spawn allowance at its 15 ms value, giving raw-error selection 60 ms for optimization raises PSNR by +3.69 [3.55, 3.83] and +2.94 [2.89, 3.01] dB on `fr2_xyz`/`fr1_xyz`, despite maps of 71k/79k. Capping the error-only map at 23k adds +2.76 [2.62, 2.90] and +1.00 [0.96, 1.05] dB, but multiplies depth L1 by 2.8 on `fr1_xyz`. A 39k cap, close to that scene's coverage map size, adds +0.04 [−0.03, 0.10] dB there (and +0.80 [0.67, 0.93] dB on `fr2_xyz`). Thus a static size limit does not reproduce coverage throttling's effect on `fr1_xyz`. For raw-error selection on `fr2_xyz`, coverage reduces the share of modeled update budget spent on Gaussians younger than ten frames from 22.7% to 9.5%, while the per-frame selection rate among mature Gaussians rises from 37.9% to 87.4%. The corresponding mature selection rate on `fr1_xyz` rises from 28.8% to 57.4%. Allocation shifts and the budget intervention support compute dilution under the tested protocol, but coverage throttling also changes the optimization trajectory (its spawn cap carries most of the effect; see the decomposition below). Map size alone is not identified as the unique cause.

Exploratory transfer to two more TUM scenes (150f×3 seeds, same six factorial cells, earlier code revision): coverage adds +0.34/+0.10 dB to error/temporal selection on `fr1_desk`, but depth L1 worsens from 0.0481/0.0476 to 0.0590/0.0595 m. On `fr3_static`, coverage adds +0.77/+0.38 dB and slightly improves depth L1. Adding backlog to temporal selection on `fr1_desk` loses 0.59 dB. These scene-level differences limit any claim that one growth rule is optimal across sensor and motion conditions; the three-seed comparisons are descriptive (full traces in `results/dilution_factorial/`).

Replica factorial and proportional budget (Tab. 3). We ran the same selector×throttle cells on two Replica scenes. Unlike TUM, coverage throttling lowers raw-error PSNR (−0.68 and −0.87 dB on `office0` and `office1`) and shrinks the map (113k to 65k and 84k to 40k); backlog throttling is worse still (−2.55 and −2.04 dB relative to coverage alone, with depth L1 up by 0.12 and 0.05 m). Without throttling, the temporal-influence selector trails raw error by 1.95 and 2.33 dB. Budget interventions help: a fixed ×4 budget adds +2.81 and +1.56 dB, and a budget proportional to map

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.85	14.75	0.4655	67.2	21,128	+8.89 ✓
error_only	24.21	20.20	0.7023	11.0	72,366	+3.53 ✓
error_influence	25.38	22.35	0.7338	11.1	86,988	+2.36 ✓
error_influence_temporal	25.23	22.05	0.7315	10.9	86,187	+2.51 ✓
rtg_slam_reimpl	23.53	20.57	0.6833	10.1	60,050	+4.21 ✓
OURS	**27.74**	**28.73**	**0.7978**	**12.1**	**21,642**	—
FULL	*27.89*	*29.29*	*0.8036*	*11.3*	*70,295*	-0.15 ✓

Table 1. **Main benchmark, TUM `fr2_xyz`, 150 frames × 8 seeds.** All six controls differ significantly from OURS after Holm correction ($p_{\text{Holm}} = 0.0469$). FULL is 0.15 dB higher. Headroom recovery is 98.3% relative to no-op and 94.0% relative to the strongest budgeted baseline, at a 3.2× smaller map than FULL. Source: commit tag `paper-rerun-1`.

	TUM fr2_xyz			TUM fr1_xyz		
Selector / map control	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	24.13	.0205	73.1	21.52	.0378	79.1
Error, coverage	27.66	.0118	21.7	23.23	.0335	37.1
Error, coverage + backlog	27.67	.0117	21.5	23.32	.0334	35.1
Temporal×influence, no throttle	25.26	.0134	86.2	22.28	.0368	85.5
Temporal×influence, coverage	27.75	.0108	21.5	23.78	.0314	37.9
Temporal×influence, coverage + backlog	27.73	.0109	21.8	23.89	.0318	33.1
Error, optimization budget ×4	27.82	.0107	71.3	24.46	.0309	79.1
Error, hard map cap 23k	26.89	.0217	23.0	22.52	.1075	23.0
Error, hard map cap 39k	24.93	.0237	39.0	21.56	.1029	39.0

Table 2. **Corrected-pose factorial and dilution controls, 150 frames × 8 seeds per scene.** All factorial cells use the same modeled 15 ms optimization budget. The ×4 control uses 60 ms for optimization while keeping the original 15 ms spawn allowance. Depth L1 is in meters. Backlog is enabled independently of warmup only in the last throttle level. Per-run traces and paired intervals are in `results/rerun_paper1/dilution_factorial/`.

size (mean scale 3.67× and 3.87×, map size unchanged) adds +2.88 and +1.56 dB. On TUM the proportional budget adds +3.73 and +2.99 dB on `fr2_xyz` and `fr1_xyz` over the unthrottled baseline measured in the same batch (24.16 and 21.54 dB). These batches use the same code revision as Tabs. 1 and 2. **Dose-response (Tab. 4).** With free growth, PSNR rises monotonically with the budget multiplier on all four scenes (for example 24.24, 26.49, 27.82 dB on `fr2_xyz`), and the fraction of mature Gaussians updated per frame rises with it; their per-seed Spearman correlation over the three free-growth cells is +1.00 in every scene. Over all six cells it stays +0.92 and +0.75 on the TUM scenes, where a 23k cap is close to the natural map size, but not on Replica (+0.20 and −0.26), where the caps (60k, 40k) remove capacity and PSNR falls by 6–11 dB regardless of budget. Under a cap, ×2 and ×4 coincide once coverage reaches 1.00, because the budget exceeds what the capped map can use. Update coverage therefore tracks quality wherever capacity is not binding. These are within-scene interventions on raw-error selection. They show that the system is compute limited and that the limit tightens as the map grows, but more compute improves any compute-limited optimizer, so they do not by themselves show that growth causes the limitation; the cap cells address that only on TUM.

Scene-level evidence (Tab. 5). Because the unit of generalization is the scene, we ran the unthrottled, coverage-throttled and map-proportional raw-error policies on 11 further scenes (six Replica, five TUM; 99 frames, three seeds) and pooled them with the four scenes above. The map-proportional budget improves PSNR on all 15 scenes (median +2.70 dB, range +0.02 to +3.73 dB; exact sign test $p < 10^{-4}$); the one near-zero gain is the small-map static scene `fr3_sitting_static` (+0.02 dB, 17k Gaussians). Coverage throttling improves PSNR on 5 of 15 scenes (median −0.13 dB, range −1.64 to +3.53 dB; sign test $p = 0.30$). The proportional-budget gain tends to grow with the unthrottled map size (Spearman +0.52, $p = 0.048$, $n = 15$, exploratory), as dilution predicts. Per-scene intervals are over three or eight seeds on a fixed trajectory, so the sign and scale across scenes are the evidence. More budget helps any compute-limited optimizer: this shows that raw-error selection under a 15 ms budget is compute limited in every scene tested, not that growth is the cause.

Longer horizon (Tab. 6). On the three TUM scenes whose sequences allow it (`fr2_xyz`, `fr1_xyz`, `fr1_desk`; 500 frames, two seeds), the unthrottled raw-error baseline degrades with horizon (window-mean PSNR

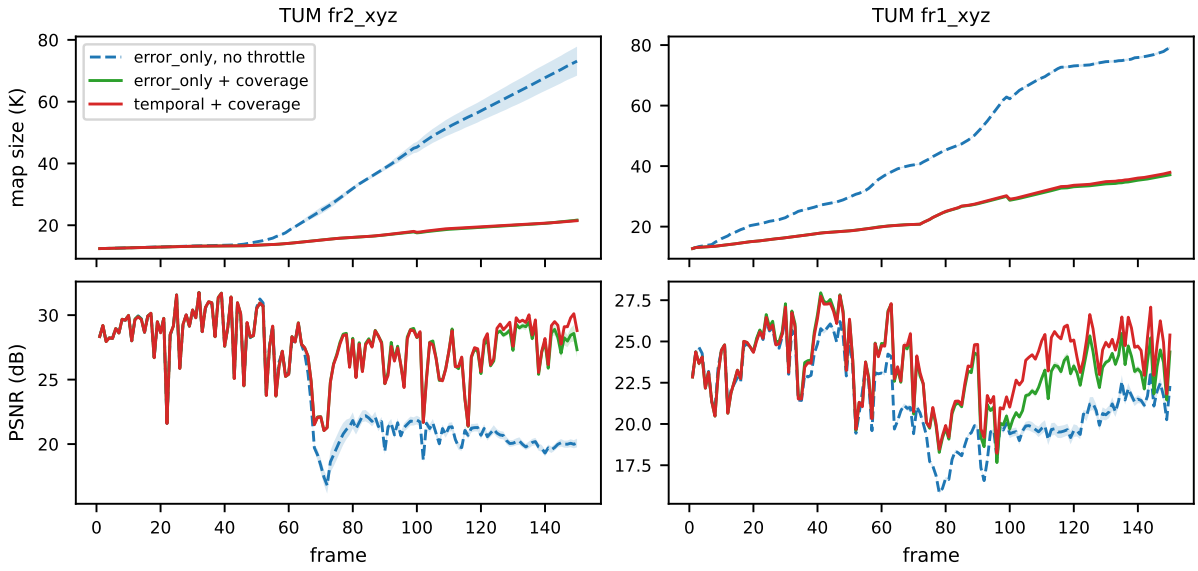


Figure 1. **Map growth and reconstruction quality over time** on TUM `fr2_xyz` and `fr1_xyz` (150 frames, eight seeds). Lines show the seed mean and shaded bands show ±1 standard deviation. `error_only` versus `error + coverage` holds the selector fixed to show the coverage effect; `temporal + coverage` adds the combined policy. Paired sequence-level contrasts are reported in Tab. 2.

	Replica `office0`			Replica `office1`		
Condition	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	30.63	.0403	112.7	33.84	.0161	84.2
Error, coverage	29.95	.0556	65.2	32.97	.0273	39.9
Error, coverage + backlog	27.40	.1720	45.7	30.93	.0744	25.9
Temporal×influence, no throttle	28.68	.0489	115.8	31.51	.0202	85.8
Temporal×influence, coverage	28.14	.0651	66.3	31.73	.0329	42.0
Temporal×influence, coverage + backlog	26.18	.2385	45.8	29.79	.1093	24.1
Error, optimization budget ×4	33.44	.0326	107.5	35.41	.0106	82.8
Error, budget ∝ map size	33.51	.0327	107.5	35.41	.0108	82.7

Table 3. **Replica factorial and map-proportional budget, 99 (`office0`) and 119 (`office1`) frames × 8 seeds per scene.** Same cells as Tab. 2; all runs use the current code revision and a 15 ms modeled budget unless stated. The proportional cell sets the optimization budget to 15 ms × N_t/N_{ref} (mean scale 3.7–3.9×) with the spawn allowance fixed at 15 ms. Coverage and backlog throttling lower PSNR on Replica, whereas more optimization budget raises it.

24.22 → 17.88, 21.50 → 19.32 and 22.49 → 16.95 dB from frames 1–150 to 301–500) and its final map reaches 116–136k Gaussians. The map-proportional budget (mean scale 5.4–6.7×) recovers +3.65 → +8.48, +3.03 → +5.77 and +1.42 → +5.92 dB over the same windows, whereas coverage throttling is not monotone (+3.47 → +4.85, +1.71 → +0.72, +0.47 → +1.17 dB). The gap to the unthrottled baseline therefore widens with horizon, consistent with dilution accumulating as the map grows, but with two seeds and three scenes this is descriptive, and the proportional budget spends several times more compute at 500 frames. Replica sequences are too short (99–119 frames) for this test.

Decomposing coverage throttling (Tab. 7). Coverage throttling combines a densification-threshold multiplier (up to 1.5×) with a spawn cap of 0.2× once coverage saturates; four seeds per scene separate them. The spawn cap alone reproduces nearly all of the effect (+3.25 vs +3.36 dB on `fr2_xyz`, +1.55 vs +1.70 on `fr1_xyz`, −0.67 vs −0.67 on `office0`), whereas the threshold multiplier alone gives +1.12, +0.18 and +0.02 dB. A 0.2× spawn cap applied whenever the view is settled, without the coverage gate, matches the full policy on TUM (+3.30, +1.54 dB) and is slightly worse on Replica (−0.72, −1.14 dB); a 0.5× cap gives intermediate results (+1.56, +0.54, −0.11, −0.11 dB). The coverage gate therefore adds little beyond limiting the growth rate. Adding throttling to the map-proportional budget adds nothing on TUM (+3.67 vs +3.69 dB on `fr2_xyz`) and costs about 1.6 dB on both Replica scenes relative to the proportional budget alone (+1.32 vs +2.89 and −0.08

Scene (cap)	Free map growth ×1	Free map growth ×2	Free map growth ×4	Hard map cap ×1	Hard map cap ×2	Hard map cap ×4	ρ_{all}	ρ_{free}
`fr2_xyz` (23k)	24.24 (0.38)	26.49 (0.58)	27.82 (0.98)	26.90 (0.73)	27.85 (1.00)	27.85 (1.00)	+0.92	+1.00
`fr1_xyz` (23k)	21.53 (0.29)	22.78 (0.55)	24.45 (0.89)	22.52 (0.65)	23.63 (1.00)	23.63 (1.00)	+0.75	+1.00
`office0` (60k)	30.61 (0.17)	32.25 (0.34)	33.43 (0.72)	24.37 (0.24)	25.25 (0.49)	26.07 (1.00)	+0.20	+1.00
`office1` (40k)	33.85 (0.18)	34.88 (0.33)	35.42 (0.68)	22.79 (0.38)	22.85 (0.75)	22.90 (1.00)	-0.26	+1.00

Table 4. **Dose-response in optimization budget, 8 seeds.** Raw-error selection, no throttling; entries are mean PSNR (dB) with the mature-update coverage in parentheses, for a budget of $15 \times \{1, 2, 4\}$ ms with the spawn allowance fixed at 15 ms, with free growth or a hard Gaussian cap. ρ is the per-seed Spearman correlation between PSNR and mature-update coverage over the cells (all six, or the three free-growth cells), averaged over seeds. Once coverage reaches 1.00 extra budget is unused and PSNR saturates. On Replica the caps remove needed capacity (PSNR 6–11 dB lower at equal budget), so the cross-regime correlation does not hold there.

Scene	Prop.	Cov.	Map (k)	Scene	Prop.	Cov.	Map (k)
`fr2_xyz` (T)	+3.73	+3.53	75	`office1` (R)	+1.56	−0.87	84
`fr1_xyz` (T)	+2.99	+1.71	79	`office2` (R)	+2.76	−0.08	115
`fr1_desk` (T)	+0.92	−0.16	60	`office3` (R)	+3.55	−0.26	148
`fr1_room` (T)	+1.74	−0.13	103	`office4` (R)	+3.68	−0.04	135
`fr1_rpy` (T)	+2.06	+0.04	72	`room0` (R)	+2.54	−0.19	149
`fr3_sitting_static` (T)	+0.02	+0.08	17	`room1` (R)	+2.12	−0.25	102
`fr3_structure_texture_near` (T)	+2.70	+2.54	61	`room2` (R)	+3.09	−1.64	145
`office0` (R)	+2.88	−0.68	113				

Table 5. **Scene-level change in mean PSNR (dB) relative to unthrottled raw-error selection, 15 scenes.** Prop.: optimization budget proportional to map size; Cov.: coverage throttling; Map: final size of the unthrottled map. T: TUM, R: Replica. `fr2_xyz` and `fr1_xyz` use 150 frames, `office0` and `office1` use 99 and 119 frames, all with eight seeds; the others use 99 frames and three seeds. Proportional budget is positive on 15 of 15 scenes (median +2.70, exact sign test $p < 10^{-4}$); coverage throttling on 5 of 15 (median −0.13, $p = 0.30$). Spearman correlation between map size and proportional-budget gain: +0.52 ($p = 0.048$, exploratory).

Scene	Frames	Base	Cov.	Prop.
`fr2_xyz`	1–150	24.22	+3.47	+3.65
	151–300	18.70	+6.07	+9.30
	301–500	17.88	+4.85	+8.48
`fr1_xyz`	1–150	21.50	+1.71	+3.03
	151–300	19.74	+1.88	+5.37
	301–500	19.32	+0.72	+5.77
`fr1_desk`	1–150	22.49	+0.47	+1.42
	151–300	18.98	+2.52	+5.05
	301–500	16.95	+1.17	+5.92

Table 6. **Long horizon (500 frames, two seeds).** Mean PSNR of the unthrottled raw-error baseline (Base) and the change from coverage throttling (Cov.) and the map-proportional budget (Prop., mean scale 5.4–6.7×) in three frame windows. The baseline degrades with horizon and the gains grow; coverage throttling is not monotone. Replica sequences are too short for this test.

Condition	`fr2_xyz`	`fr1_xyz`	`office0`	`office1`
Coverage throttling (full)	+3.36	+1.70	−0.67	−0.88
spawn cap only	+3.25	+1.55	−0.67	−0.91
threshold only	+1.12	+0.18	+0.02	−0.01
Spawn cap ×0.2, ungated	+3.30	+1.54	−0.72	−1.14
Spawn cap ×0.5, ungated	+1.56	+0.54	−0.11	−0.11
Budget ∝ map size	+3.69	+2.96	+2.89	+1.54
Throttling + budget ∝ map	+3.67	+2.82	+1.32	−0.08

Table 7. **Decomposition of coverage throttling, change in mean PSNR (dB) relative to unthrottled raw-error selection, four seeds.** Spawn-cap-only fixes the densification-threshold multiplier at 1; threshold-only disables the spawn cap; the ungated caps apply whenever the view is settled. Budget ∝ map size uses a mean 3.7–6× larger optimization budget than the other rows.

vs +1.54 dB). The proportional budget spends several times more compute than the fixed-budget rows, so the two controls address different constraints: capping growth when compute is fixed, scaling the budget when it can grow.

Held-out views (Tab. 8). Every eighth frame is rendered but never mapped or optimized. The sign of every effect is unchanged on these frames but its size shrinks: the proportional budget's gain is +2.18, +1.69, +1.40 and +0.68 dB held out versus +2.85, +2.59, +2.58 and +1.44 dB on mapped frames (47–76% retained), and the train-minus-held-out gap grows with the budget (for example 3.29 → 4.47 dB on `office0`). Throttling's held-out changes are +2.13 and +1.04 dB on TUM and −0.27 and −0.57 dB on Replica. Part of the gain from more optimization is therefore specific to the mapped views. The held-out frames lie on the mapped trajectory, so this proxy does not test generalization to new trajectories.

Quality at measured cost (Tab. 9). Quality per unit of

Mapped / held-out	fr2_xyz	fr1_xyz	office0	office1
Coverage throttling	+2.73/+2.13	+1.50/+1.04	−0.62/−0.27	−0.93/−0.57
Spawn cap ×0.2	+2.68/+2.11	+1.38/+0.96	−0.77/−0.50	−1.08/−0.54
Budget ∝ map	+2.85/+2.18	+2.59/+1.69	+2.58/+1.40	+1.44/+0.68
Gap, unthrottled	0.87	0.98	3.29	2.79
Gap, budget ∝ map	1.53	1.87	4.47	3.55

Table 8. **Mapped versus held-out views, change in mean PSNR (dB) relative to unthrottled raw-error selection, four seeds.** Every eighth frame is rendered but never mapped (a same-trajectory proxy, not an independent trajectory). The last rows give the train-minus-held-out PSNR gap.

compute differs from quality per modeled budget, because the map size changes the rendering cost. We sweep the modeled budget (×1, ×2, ×4) with free growth or a ×0.2 spawn cap (four seeds). The proportional-budget gains cost real time: the mean end-to-end frame time rises from 86 to 135 ms on `office0`. On the TUM scenes the capped map is both faster and as good or better: on `fr2_xyz` the capped ×1 cell reaches 27.53 dB in 79 ms, versus 24.08 dB in 88 ms for free growth and 27.81 dB in 96 ms for free growth at ×4; on `fr1_xyz` the capped ×2 cell reaches 24.40 dB in 87 ms versus 24.45 dB in 119 ms for free growth at ×4. On Replica, free growth with more budget is better at equal time: interpolating its curve at the capped cells' frame time gives 0.36–0.64 dB (`office0`) and 0.96–1.35 dB (`office1`) more PSNR than the capped ×2 and ×4 cells. Timings come from one laptop GPU and indicate relative costs only. A spawn cap is thus the better use of compute on TUM and the worse on Replica, matching the modeled-budget result, and the proportional budget's gains are paid for in wall-clock time. **Estimated poses (Tab. 10).** We also map with poses from a frame-to-frame dense RGB-D Gauss-Newton tracker (ATE 1.24 and 2.45 cm; it does not use the map, so ATE is shared across conditions). PSNR at the tracked pose improves by +4.09 to +4.44 dB on `fr2_xyz` and +3.12 to +4.48 dB on `fr1_xyz`, more than with ground-truth poses. Rendering the same maps from the ground-truth pose removes almost all of this (+0.02 to +0.03 and +0.15 to +0.22 dB; baseline 21.88 and 17.58 dB versus 27.35 and 23.75 dB at the tracked pose), because pose error rather than optimization budget then limits the alignment. Gains in self-consistent quality therefore do not imply better alignment with the true camera. In a map-based SLAM system a better map could also improve tracking, which this map-independent tracker cannot show.

Observed failure cases. Two TUM scenes show where the combined policy trails raw-error selection. *Exploration-heavy motion* (`fr1_room`, −1.04 dB) coincides with a smaller final map (92k vs 97k); the available run does not isolate whether throttling or selection causes the gap. *Rapid rotation* (`fr1_rpy`, −1.13 dB mean, finals 24.5 vs 24.1) is consistent with a lag from temporal smoothing, but this attribution has not been tested independently. **Horizon effect (Fig. 1):** at 30 frames OURS ties `error_only` (−0.01 dB on `fr2_xyz`); the gap appears at longer horizons. The new age-group logs in Tab. 2 link this divergence to a smaller update share for mature Gaussians in the growing map. Around frame 65 in seed 42, both error-only variants encounter difficult views; the unthrottled run continues spawning roughly 540–720 splats per frame and its PSNR remains low after frame 70, whereas the coverage run spawns roughly 60–130 (mean 92) and recovers by frame 75. The unthrottled loss rises from about 0.07 at frame 65 to 0.13 at frame 70 and stays high; opacity declines while the map expands. This is sustained divergence following shared hard views, not evidence for a single isolated gradient explosion. The logs cannot rule out floaters without a spatial geometry audit.

Scene dependence (Supp. Tab. S7). The combined policy wins the two confirmatory TUM comparisons ($n = 8$): `fr2_xyz` +3.53 dB and `fr1_xyz` +2.38 dB over raw-error selection, both $p = 0.0078$. It trails raw-error selection on all eight Replica scenes (−1.93 to −4.21 dB; `office0` and `office1` at $n = 8$, the rest exploratory at $n = 3$) and on `fr1_room` and `fr1_rpy`, so across 14 sequences it loses mean PSNR on 10. On Replica `office0`, raw-error selection with a weak throttle ($\kappa = 0.98$) reaches 30.17 dB versus 26.85 dB for the combined policy (supplement). The Replica factorial (Tab. 3) locates the loss: the unthrottled raw-error cell is the best of the six factorial cells on both scenes, adding the temporal-influence selector costs 1.95 and 2.33 dB, and coverage throttling costs 0.68 and 0.87 dB for raw-error selection. We cannot tell whether this reflects Replica's rendering, trajectories or depth quality. Compute allocation still matters there, since a map-proportional budget adds +2.88 and +1.56 dB. A simple hole-rate router that picks the better arm on the two datasets is in the supplement; its within-scene depth interventions did not reverse the ranking, so we do not use it as evidence for any mechanism.

Systems. Scheduler overhead is 2.54 ms; optimization-budgeted execution runs at about 10 FPS, so this is *not* end-to-end 15 ms real time. A separate baseline odometer (dense photometric+depth Gauss-Newton, no keyframes or loop closure) reaches ATE 0.9/2.0 cm on `fr2_xyz`/`fr1_xyz` over 100 frames (Supp. Tab. S3); it is a sanity check, not a SLAM claim. AlexNet LPIPS on `fr2_xyz` (150f×3) is 0.211 for OURS versus 0.294 for `error_only` (earlier code revision); depth L1 is in Tab. 2, and we report no mesh-based surface accuracy. Published-system numbers use different frames, resolution and views and give range context only (Supp. Tab. S2); short-horizon budget-curve and coverage-knee ablations are in Supp. Figs. S1–S2. **Reproducibility.** Every table maps to a frozen bundle and a command in `paper_repro/`; the headline bundle records per-policy configs, source hashes, artifact hashes,

Scene	Free map growth ×1	Free map growth ×2	Free map growth ×4	Spawn cap ×0.2 ×1	Spawn cap ×0.2 ×2	Spawn cap ×0.2 ×4
fr2_xyz	24.08 (88)	26.50 (96)	27.81 (96)	27.53 (79)	27.92 (62)	27.93 (63)
fr1_xyz	21.51 (100)	22.77 (108)	24.45 (119)	23.10 (92)	24.40 (87)	24.44 (76)
office0	30.62 (86)	32.26 (105)	33.44 (135)	29.89 (79)	31.31 (98)	31.77 (109)
office1	33.85 (86)	34.86 (107)	35.42 (138)	32.74 (80)	33.48 (98)	33.53 (107)

Table 9. **Quality at measured cost.** Mean PSNR (dB) with the mean end-to-end frame time in ms in parentheses, for a modeled optimization budget of $15 \times \{1, 2, 4\}$ ms (spawn allowance fixed) with free growth or a $\times 0.2$ spawn cap; raw-error selection, four seeds. Timings come from one laptop GPU and indicate relative costs only.

Scene	Condition	Tracked pose	GT pose
fr2_xyz (ATE 1.24 cm)	unthrottled	27.35	21.88
	coverage	+4.31	+0.02
	spawn cap ×0.2	+4.09	+0.03
	budget ∝ map	+4.44	+0.02
fr1_xyz (ATE 2.45 cm)	unthrottled	23.75	17.58
	coverage	+3.40	+0.15
	spawn cap ×0.2	+3.12	+0.15
	budget ∝ map	+4.48	+0.22

Table 10. **Mapping with estimated poses, three seeds.** PSNR (dB) of the unthrottled baseline and the change for each condition, evaluated at the tracker's pose and when the same map renders from the ground-truth pose. The tracker is a frame-to-frame RGB-D Gauss-Newton odometer that does not use the map.

pose protocol and clean-worktree status of each run set. Prior pose-bugged numbers (19.24 dB) are quarantined in `results/final_confirmation/DEPRECATED.md` and never compared. Environment: RTX 4050 Laptop GPU (6 GB), PyTorch 2.11+cu128, `gsplat` [20] 1.5.3, Python 3.12; peak allocation stays under 3.5 GB with the desktop compositor active, which is the intended deployment envelope for budgeted online mapping.

5. Limitations

Scope of the mechanism. The proportional-budget contrast covers 15 scenes (11 with three seeds and 99 frames), the factorial and dose-response four scenes; each uses one trajectory per scene and raw-error selection only. Coverage throttling helps on TUM and hurts on Replica; its spawn cap carries nearly all of the effect (Tab. 7, four seeds), while map caps neither isolate map size nor remain valid on Replica, where they remove needed capacity. More budget improves any compute-limited optimizer, so the budget interventions show a compute limitation that tightens with map size, not that growth itself is harmful. Tabs. 1, 2, 3 and 4 share one code revision; Supp. Tab. S7 and the supplementary tables come from earlier revisions. Additional TUM scenes are exploratory ($n = 3$) and show smaller or depth-degrading effects. **Protocol.** Mapping uses ground-truth poses in the main experiments (estimated-pose results are in Tab. 10, with a map-independent tracker); main tables use up to 150 frames (about five seconds at 30 FPS) and 500-frame runs cover three TUM scenes with two seeds; PSNR is mostly measured on mapped views, with a same-trajectory held-out proxy (Tab. 8) that retains 47–76% of the proportional-budget gain. The 15 ms budget is modeled optimization time, not frame latency, and the pipeline runs at about 8–11 FPS on a laptop GPU. We do not compare against offline density-control methods [1, 7, 9, 22] or integrate the policy into a full SLAM system with its own tracker. **Statistics.** Eight seeds establish repeatability on fixed trajectories; the unit of generalization is the scene, and only two scenes are confirmatory. **Open items.** Longer sequences with a real tracker, comparisons against density-control baselines, additional scenes and sensors, and capacity-matched cap controls on Replica would test whether dilution generalizes beyond these four scenes.

6. Conclusion

Under a fixed modeled optimization budget, a growing map leaves established Gaussians under-updated. Raising the budget in proportion to map size improves raw-error selection on all 15 scenes tested (median +2.70 dB), PSNR rises monotonically with budget, and the fraction of mature Gaussians updated per frame tracks quality wherever map capacity is not binding. The remedy of throttling spawns is scene dependent: it improves 5 of 15 scenes, recovering several dB on two TUM scenes (27.74 dB on `fr2_xyz`, 0.15 dB below unconstrained optimization at a 3.2× smaller map) but lowers PSNR on Replica, where the combined policy trails raw-error selection. We therefore present dilution as a compute-allocation mechanism with scene-dependent remedies, and leave open whether it persists on longer sequences and inside full SLAM systems.

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